

We tend to underestimate the medical skills of apparently undeveloped communities—particularly during the so-called Dark Ages in medieval Europe—but it is evident that many people had a surprisingly sophisticated understanding of plant medicine. For example, recent excavations at an 11th-century monastic hospital in Scotland revealed that the monks were using exotic herbs such as opium poppy (*Papaver somniferum*, p. 244) and marijuana (*Cannabis sativa*, p. 182) as painkillers and anesthetics. Likewise, the herbalists in Myddfai, a village in South Wales, obviously knew of Hippocrates' writings in the 6th century CE and used a wide variety of medicinal plants. The texts that have been handed down from that herbal tradition are filled with an engaging blend of superstition and wisdom. Two prescriptions from a 13th-century manuscript illustrate the point. The first recipe could have been written by a modern, scientifically trained herbalist; the second, one must presume, is pure fancy, and would not choose to try it out!

To Strengthen the Sight

Take Eyebright and Red Fennel, a handful of each, and half a handful of Rue, distil, and wash your eye daily therewith.

To Destroy a Worm in the Tooth

Take the root of a cat's ear, bruise, and apply to the patient's tooth for three nights, and it will kill the worm.

Islamic & Indian Medicine 500–1500 CE

Folk medicine was largely unaffected by sweeping forces of history, but Western scholastic medicine suffered greatly with the decline of the Roman Empire.

It was thanks to the flowering of Arabic culture in 500–1300 CE that the gains of the classical Greek and

Roman period were preserved and elaborated. The spread of Islamic culture along North Africa and into present-day Italy, Spain, and Portugal led to the founding of renowned medical schools, notably at Cordoba in Spain. The Arabs were expert pharmacists, blending and mixing herbs to improve their medicinal effect and their taste. Their contacts with both Indian and Chinese medical traditions meant that they had a remarkable range of medical and herbal knowledge to draw on and develop. Avicenna (980–1037 CE), author of *Canon of Medicine*, was the most famous physician of the day, but perhaps the most unusual herbal connection was made a century before his time by Ibn Cordoba, an intrepid Arab seafarer, who brought ginseng root (*Panax ginseng*, p. 118) from China to Europe. This valuable tonic herb was to be regularly imported into Europe from the 16th century onward.

Further east, in India, the 7th century saw a golden age of medicine. Thousands of students studied Ayurveda at university, especially at Nalanda. There, scholars recorded the medical achievements of the time, with advances such as the development of hospitals, maternity homes, and the planting of medicinal herb gardens.

Central & South American Cures

On the other side of the world, the ancient civilizations of Central and South America—Maya, Aztec, and Inca—all had herbal traditions with a profound understanding of local medicinal plants. One account tells of Incas taking local herbalists from what is now Bolivia back to their capital Cuzco in Peru because of the herbalists' great capabilities, which reputedly included growing penicillin on green banana skins.

At the same time, medicine and religion were still closely interwoven in these cultures, possibly even more so than in Europe. One gruesome account tells of Aztec sufferers of skin diseases who sought to appease the god Xipe Totec by wearing the flayed skins of sacrificial victims. Fortunately, a supernatural appeal to the gods was not the sole means to relieve this and other afflictions. Many herbs were available as alternative treatments, including sarsaparilla (*Smilax* spp., p. 270), a tonic and cleansing herb that was used in treatments for a variety of skin complaints including eczema and psoriasis.



Galen and Hippocrates, two of the preeminent physicians of the classical era, debate in this imaginary scene depicted in a fresco.